

SkillAnchor: An Auditable LLM Framework for Job-Skill Span Extraction With Supervised Fine-Tuning and Multi-Source Retrieval

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ABSTRACT Exact job-skill span extraction converts free-text job advertisements into token-aligned skill mentions for labor-market analysis and matching applications. Generative large language models (LLMs) offer a flexible extraction interface, but their outputs must still satisfy exact span boundaries and a reproducible BIO evaluation contract. We present SkillAnchor, an open-weight, artifact-mapped framework for auditable generative extraction. Its reported path, SkillAnchor-Gen, combines span-anchored LoRA supervised fine-tuning, retrieval of in-domain training exemplars and ESCO definitions, dataset-specific prompting, and BIO-only decoding. We evaluate this fixed path on documented processed splits from six public corpora, alongside matched XLM-R and GLiNER baselines and traceable component controls. XLM-R remains the strongest aggregate system, while SkillAnchor-Gen is competitive on SkillSpan and Kompetenzer. Operational SFT controls and source-isolated retrieval decodes further show that neither adaptation nor retrieval has a corpus-invariant effect: concatenating training examples and taxonomy definitions can be complementary on some corpora and destructive on others. SkillAnchor therefore contributes a reproducible generative baseline and cross-corpus evidence about heterogeneous retrieval sources, rather than a claim of uniform superiority over supervised extraction.

INDEX TERMS Large language models, retrieval-augmented generation, skill extraction, named entity recognition, BIO tagging.

I. INTRODUCTION

JOB-skill extraction identifies exact skill mentions in job advertisements for labor-market analysis, vacancy-candidate matching, and taxonomy mapping. In the corpora studied here, each prediction must correspond to a contiguous token span and is evaluated under a BIO exact-match contract [1], [2]. This requirement distinguishes the task from unconstrained phrase generation: a plausible skill that is absent from the sentence, or a boundary that includes one extra token, is still an error. The central challenge is therefore to retain the flexibility of generative LLMs while producing reproducible, sentence-grounded spans.

Three method families frame this challenge. Supervised sequence taggers built on multilingual encoders such as

XLM-R remain strong in-domain baselines [2], [3]. Generalist extractors such as GLiNER and UniversalNER broaden label coverage, whereas generative IE systems provide a flexible output interface but do not inherently guarantee token-aligned boundaries [4]–[8]. Retrieval augmentation can supply demonstrations and external definitions [9], but irrelevant or poorly placed context can also distract an LLM [10]–[13]. For job-skill extraction, the central question is whether heterogeneous evidence sources can be safely combined under exact-span evaluation.

We present **SkillAnchor**, an open-weight, artifact-mapped framework for studying these choices (Fig. 1). Its reported generative method, SkillAnchor-Gen, has four parts. First, span-anchored LoRA SFT adapts Qwen2.5-14B-Instruct to

SkillAnchor: Auditable Job-Skill Span Extraction

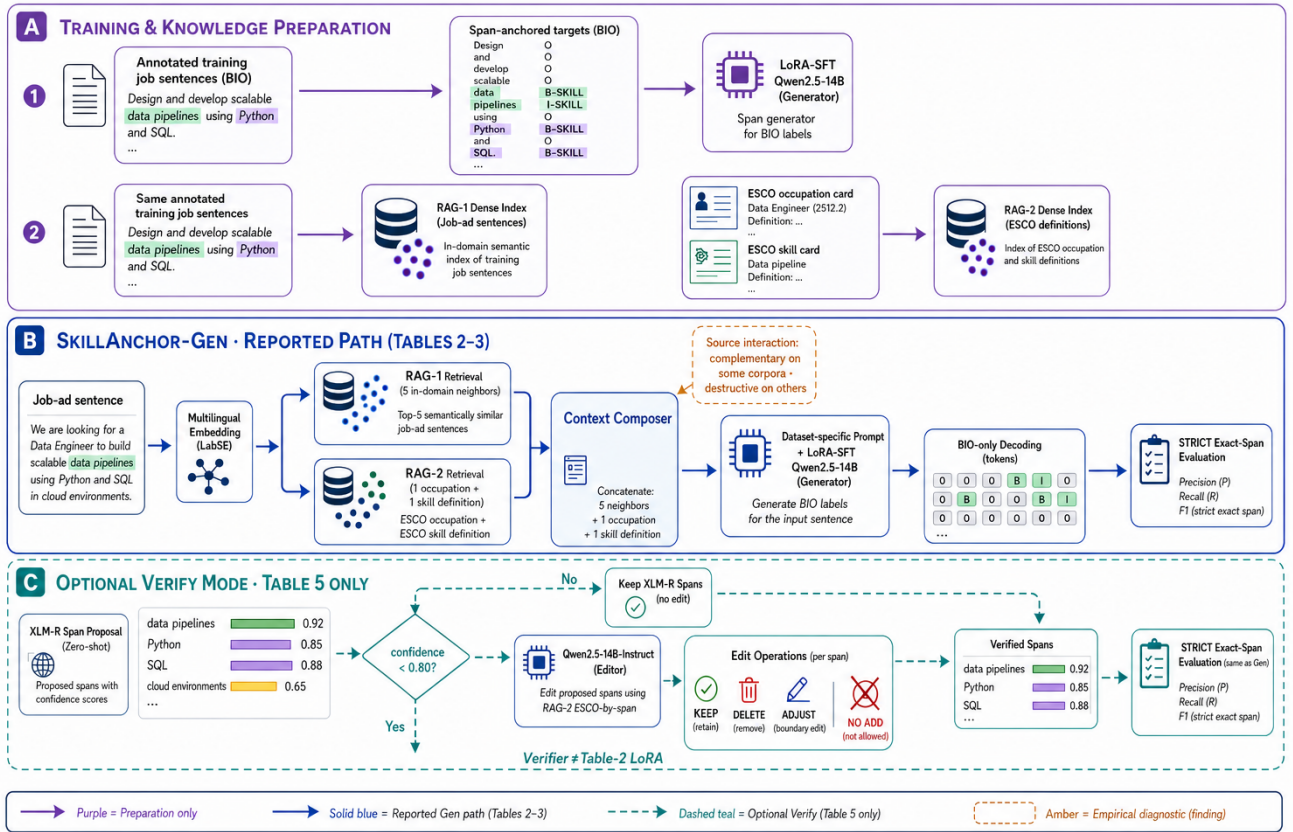


FIGURE 1: Overview of SkillAnchor. Panels A–B show the preparation and reported SkillAnchor-Gen path for multi-source retrieval, BIO-only decoding, and STRICT evaluation (Tables 2–3). Panel C is an auxiliary Verify diagnostic outside the reported generative method (Table 5).

emit BIO sequences rather than free-form skill lists. Second, dual-source retrieval supplies semantically similar annotated training examples and ESCO occupation and skill definitions [14]. Third, a dataset-specific prompt states the language, annotation convention, and output contract. Finally, BIO-only decoding restricts the response interface so that every prediction can be scored by the same exact-span evaluator. This fixed path is the sole SkillAnchor method reported in Table 2; it is not selected separately for each corpus.

A BIO-legality checker, corpus-level retrieval routing, and an XLM-R–Qwen Verify procedure are evaluated only as auxiliary diagnostics. None changes the fixed SkillAnchor-Gen configuration or its Table 2 results.

We evaluate documented processed splits derived from six public corpora—SkillSpan, Kompetenzer, Green, FIJO, Sayfullina, and Gnehm—using STRICT sequeval exact-span precision, recall, and F1 [1], [15]–[19]. XLM-R and GLiNER are scored on the same processed test splits and cleaned gold representation as SkillAnchor. Stored predictions, prompts, adapter mappings, and scorer inputs are tracked so that the matched comparison and the reported component cells can be regenerated. Earlier prompting aggregates are retained in Appendix Table 12 for transparency but are excluded from

the artifact-matched ranking.

The results support a diagnostic rather than a state-of-the-art claim. SkillAnchor-Gen is close to XLM-R on SkillSpan and leads the matched comparison only on Kompetenzer; XLM-R remains strongest in aggregate. SFT is essential on SkillSpan and Kompetenzer, but its effect is not uniform across corpora. Retrieval is even more conditional: concatenating RAG-1 and RAG-2 is stronger than either source alone on SkillSpan, Kompetenzer, and FIJO, whereas a single source is substantially stronger on Green, Sayfullina, and Gnehm. Together, these findings show why heterogeneous context should be diagnosed by source and corpus rather than assumed to be monotonically beneficial.

The contributions are as follows.

- An open-weight, artifact-mapped generative pipeline that integrates span-anchored LoRA SFT, training-example retrieval, ESCO definition retrieval, dataset-specific prompting, and BIO-only decoding under one exact-span evaluation contract.
- A matched six-corpus evaluation against XLM-R and GLiNER, with operational SFT and retrieval controls that establish the strengths and limits of the fixed SkillAnchor-Gen configuration.

- A source-isolated retrieval analysis that exposes both complementary and destructive interactions between in-domain examples and taxonomy definitions, supported by mapped prediction artifacts and a regenerable exact-span scorer.

II. PROPOSED METHOD

A. PROBLEM FORMULATION

Given a job-ad sentence $x = (w_1, \dots, w_T)$, the goal is to assign span-level BIO tags $y \in \{B, I, O\}$. Outputs must be boundary-accurate and BIO-legal. Unsupported skill mentions (spans not licensed by the sentence, and not treated here as an ESCO-membership test) remain a residual error type under STRICT matching; we do not equate every STRICT false positive with hallucination.

B. OVERVIEW

The reported SkillAnchor configuration comprises three modules (Fig. 1): (i) **SFT with span anchoring** on an open-weight LLM; (ii) **dataset-specific prompting** with BIO-only decoding; and (iii) **multi-source retrieval (RAG)** from in-domain training sentences and authoritative skill/occupation definitions. This fixed SkillAnchor-Gen path is the only proposed method reported in Table 2. Checker, routing, and Verify variants are auxiliary analyses rather than additional SkillAnchor configurations.

C. SFT WITH SPAN ANCHORING

We convert BIO labels to anchored targets by bracketing each gold span with @@...##, producing (Sentence : x , Target : $\tilde{x}$) for instruction-style SFT (LoRA on Qwen2.5-14B). To harden the model, we use (1) span-balanced sampling to reduce head-skill bias; (2) connector normalization (e.g., “A/B”, “X, Y and Z”) for cleaner boundaries; and (3) hard negatives (all-O sentences). These choices target format stability and boundary noise without enlarging the backbone; they are not a claim that SFT is optimal on every corpus.

D. DATASET-SPECIFIC PROMPTING AND BIO-ONLY DECODING

Each dataset uses a tailored prompt encoding language, domain jargon, and typical span shape. A fixed `system` segment defines the task and skill notion (aligned with ESCO); the `user` segment supplies the raw sentence (or a token list). Decoding is restricted to BIO tags so that the interface stays comparable across datasets and does not depend on free-form tool calls.

E. MULTI-SOURCE RETRIEVAL (RAG)

For each x , we retrieve:

- **RAG-1:** the top five in-domain annotated training sentences as few-shot demonstrations (phrase and boundary style matching);
- **RAG-2:** the top-ranked ESCO occupation entry and top-ranked ESCO skill entry for contextual disambiguation.

TABLE 1: Corpus snapshot. #Train and #Test are processed instance counts after the documented length filtering, not raw dataset-card totals; #Train-Skill and #Test-Skill are distinct skill strings in each processed split.

Dataset	#Train	#Test	#Train-Skill	#Test-Skill
SkillSpan	4782	3569	1805	984
Kompetencer	778	262	365	102
Sayfullina	3705	1851	803	580
Gnehm	19824	2550	4262	674
Green	8668	335	8713	582
FIJO	399	50	515	122

Both sources are dense-indexed; candidates are selected by cosine similarity $s(q, d) = \langle f(x), f(d) \rangle$ and inserted as compact snippets. The index is built from the processed training split plus ESCO definitions; test sentences are not intentionally indexed. Reported SkillAnchor runs use the frozen dual-source context construction described above, documented in the supplementary configuration; they do not use a learned gating or source-weighting formula. RAG-2 supplies occupation and skill text as context for span generation; it does not emit ESCO skill IDs or occupation labels.

F. AUXILIARY BIO-LEGALITY DIAGNOSTIC

When enabled, a deterministic checker validates BIO legality: no I after O, no crossing or nested spans, and tags restricted to $\{B, I, O\}$. On failure, we re-prompt with a legality hint under a small retry budget $K \in \{1, 3\}$. This component repairs format errors; it is not a factuality or ESCO-membership oracle, and it is not a substitute for SFT or RAG. We do not treat “no token overlap with gold” (STRICT false positives) as a measured hallucination rate until that quantity is tabulated from the stored predictions. The reported SkillAnchor configuration does not call the checker; A1–A4 do. The Gnehm A1–A4 runs use a different adapter from the SkillAnchor row (`sft_gnehm_ner` vs. `sft_gnehm_ner_new`).

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. EXPERIMENTAL SETUP

1) Datasets and Metrics

We evaluate on six public datasets spanning multiple languages and domains: *SkillSpan* [1], *Kompetencer* [15], *Green* [16], *FIJO* [17], *Sayfullina* [18], and *Gnehm* [19]. Train/test sizes and distinct skill counts are listed in Table 1. This paper does not include Chinese-language skill data.

The primary metric is span-level precision (P), recall (R), and F1 under STRICT matching, implemented as seqeval exact-span scores and reported in percent. We do not call these scores token-level: seqeval exact-span is an entity/span metric. Skill-level (partial-overlap) scores appear only as an auxiliary view on the A1–A4 table. We do not report a RELAX block.

2) Comparison Setting

All systems in the main table are scored against the same cleaned skill-span gold on the documented processed test split of each corpus. Unless noted, **SkillAnchor** denotes the fixed SkillAnchor-Gen method: LoRA SFT, dual-source retrieval from training examples and ESCO definitions, a dataset-specific prompt, and BIO-only decoding. Checker, routing, A1–A4, and Verify results are auxiliary diagnostics; they do not alter the Table 2 SkillAnchor row.

3) Baselines

XLm-R large is trained as a corpus-specific supervised token classifier; GLiNER is fine-tuned per corpus from `gliner-multi-v2.1` and converted to untyped *B/I* tags before scoring. Both baselines use the same processed splits and gold representation as SkillAnchor. Table 2 is restricted to artifact-verified XLm-R, GLiNER, and SkillAnchor results. Appendix Table 12 preserves GPT-3.5 and Qwen 5-shot aggregates from the prior manuscript verbatim. Because their associated prediction artifacts are unavailable in the current repository, they are excluded from bolding, matched ranking, and aggregate comparisons. A previously circulated Llama-3-8B row and the RELAX block remain omitted because several cells did not correspond to independent predictions.

B. MAIN RESULTS

Table 2 lists span-level STRICT P/R/F1 on the documented processed test split of each corpus.

Matched comparator results. XLm-R has the strongest unweighted six-corpus mean F1 (59.26), followed by GLiNER (48.80) and SkillAnchor (37.19). Because that mean weights FIJO ($N=50$) equally with SkillSpan ($N=3569$), Table 2 also reports test-size-weighted means of 71.77, 53.33, and 42.17, respectively. Neither summary is a deployment score or a field-wide ranking.

Appendix Table 12 is informative but not artifact-matched. In particular, the previously reported GPT-3.5 5-shot result reaches 54.91 F1 on Gnehm, above SkillAnchor at 41.29. We retain this unfavorable comparison for transparency, while avoiding a stronger claim because the GPT prediction file cannot be independently re-scored in this revision.

SkillAnchor leads the matched comparison only on Danish *Kompetencer*, reaching 43.84 F1 versus 43.08 for XLm-R and 29.79 for GLiNER. On *SkillSpan*, SkillAnchor reaches 53.38, close to but below XLm-R (55.33). GLiNER is highest on *FIJO* (44.75), while XLm-R is highest on *Green*, *Sayfullina*, and *Gnehm* (43.07, 91.03, and 88.24). These gaps show that the current 14B generative configuration does not replace a strong supervised encoder; its value lies in an explicit, ablatable interface for studying adaptation and retrieval.

C. ABLATION STUDY

We compare operational SFT and retrieval-disabled pipeline variants on Qwen2.5-14B under STRICT F1 using the same documented processed-test splits and scorer (Table 3). The

without-SFT row uses the base backbone with the frozen retrieval configuration. The without-retrieval row uses the SkillAnchor adapter but the repository's non-RAG prompt template as well as no retrieved context; it is therefore not a context-only intervention that holds prompt wording fixed. Prompt, CoT, and archived retrieval-width variants are reported separately in Appendix B; they are not used for a causal attribution to SFT or retrieved context alone. Fig. 2 plots the same STRICT F1 cells as Table 3.

SFT is dataset-dependent. Removing SFT yields the largest drop on SkillSpan (53.38 $\rightarrow$ 18.71) and Kompetencer (43.84 $\rightarrow$ 11.56), so the adapter is essential on those corpora. On Gnehm and Sayfullina, the base model with RAG exceeds the Table 2 SFT adapter (41.29 $\rightarrow$ 56.19 and 25.13 $\rightarrow$ 44.12). The prediction files are distinct from the SkillAnchor files, but a single ablation cannot identify whether the reversal is caused by adapter quality, retrieval content, or their interaction.

The retrieval-disabled pipeline is likewise dataset-dependent. Disabling retrieval reduces Kompetencer from 43.84 to 24.87, FIJO from 36.36 to 18.72, and SkillSpan from 53.38 to 45.09 ($N=3569$). The SkillSpan figure is not the discarded subset score 54.52 ($N=1001$). On Green, Gnehm, and especially Sayfullina, removing RAG raises STRICT F1 (23.13 $\rightarrow$ 26.68, 41.29 $\rightarrow$ 48.46, 25.13 $\rightarrow$ 76.87). The Sayfullina dump uses the same `sft_sayfullina_skills` adapter as Table 2; only 16.2% of BIO sequences match the SkillAnchor file, so the jump is not a copy of the main-row predictions. Gnehm without retrieval shares 85.8% of BIO sequences with SkillAnchor and still gains 7.17 F1. Because the retrieval-disabled runs also use the non-RAG prompt template, these differences characterize the operational pipelines rather than the isolated effect of adding context to an otherwise identical prompt.

Fixed configuration versus per-corpus selection. The locally strongest Table 3 cell is not SkillAnchor on every corpus: Gnehm peaks at the untuned backbone with RAG (56.19), Sayfullina at SFT without retrieval (76.87). We report a single pre-specified configuration—LoRA SFT, the frozen two-source RAG setting, and a dataset-specific prompt—rather than selecting, for each corpus, the variant with the highest STRICT F1. Those reversals are empirical findings. These reversals motivate conditional retrieval and stronger adaptation controls; we do not select a different configuration for each corpus after observing test F1.

Prompting and CoT. Appendix B compares dataset-agnostic versus dataset-specific prompts and CoT on the corpora with matched files. A dataset-agnostic prompt is below SkillAnchor on SkillSpan and Green, but higher on FIJO (42.15 versus 36.36). CoT does not improve the matched full- N Green/FIJO runs. We therefore omit CoT from the reported SkillAnchor configuration.

Summary of ablation findings

(i) SFT is the primary driver on SkillSpan and Kompetencer; on Gnehm the untuned backbone with RAG is stronger than

TABLE 2: Artifact-verified results under STRICT span matching (seqeval exact-span P/R/F1, %). All methods use the documented processed test splits and cleaned skill-span gold. Bold indicates the highest F1 in each dataset column.

Method	SkillSpan			Kompetencer			Green			FIJO			Sayfullina			Gnehm			AVG		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
<i>N</i> (test)	3569			262			335			50			1851			2550			—		
<i>Supervised encoder and generalist extractor (matched STRICT)</i>																					
XLM-R large	55.77	54.90	55.33	36.13	53.33	43.08	48.33	38.84	43.07	34.68	34.96	34.82	90.30	91.76	91.03	89.78	86.76	88.24	59.17	60.09	59.26
GLiNER	33.99	53.53	41.58	33.73	26.67	29.79	65.94	31.40	42.54	51.04	39.84	44.75	82.43	75.77	78.96	41.11	83.84	55.17	51.37	51.84	48.80
<i>Proposed generative method (matched processed test N)</i>																					
SkillAnchor (Qwen2.5-14B)	52.19	54.63	53.38	42.11	45.71	43.84	26.98	20.24	23.13	35.38	37.40	36.36	22.15	29.03	25.13	44.08	38.83	41.29	37.15	37.64	37.19

Best F1 in each dataset column is in bold within the artifact-verified XLM-R/GLiNER/SkillAnchor comparison. Unweighted macro-average F1 is 59.26 (XLM-R), 48.80 (GLiNER), and 37.19 (SkillAnchor); the test-size-weighted means are 71.77, 53.33, and 42.17. Neither is a deployment score or a field-wide ranking: FIJO ($N=50$) has the same weight as SkillSpan ($N=3569$) in the unweighted mean. Historical prompting results and their provenance qualifications are preserved in Appendix Table 12. XLM-R on Green, FIJO, Sayfullina, and Gnehm uses stored checkpoints; SkillSpan and Kompetencer were retrained for 10 epochs because the available checkpoints were incomplete. GLiNER is fine-tuned per corpus from `gliner-multi-v2.1`. Both comparator families are converted to untyped *B/I* before seqeval scoring. The SkillSpan SkillAnchor prediction has a mislabeled experiment field in file metadata; the prediction tags correspond to SFT, a dataset-specific prompt, and the frozen reported retrieval configuration. Gnehm uses `sft_gnehm_ner_new`; Kompetencer uses `sft_kompetencer_ner`; Sayfullina uses the full $N=1851$ skills run rather than the $N=1000$ `skills_ner` file.

TABLE 3: Operational SFT and retrieval comparisons under STRICT F1 (%). Completed cells use the documented processed test split and the same STRICT scorer. SkillAnchor is SFT + the frozen two-source 5+1+1 RAG setting + a dataset-specific prompt. Without SFT uses the base 14B backbone with the same retrieval index. Without retrieval uses the SkillAnchor adapter with the non-RAG prompt template and no retrieved context; it is not a context-only intervention. Prompt / CoT / archived retrieval-width variants are in Appendix B and are not used for a causal attribution to SFT or retrieved context alone.

Setting	SkillSpan	Kompetencer	Green	FIJO	Sayfullina	Gnehm
<i>N</i>	3569	262	335	50	1851	2550
SkillAnchor (SFT+RAG)	53.38	43.84	23.13	36.36	25.13	41.29
Without SFT (base + retrieval)	18.71	11.56	20.33	26.52	44.12	56.19
Without retrieval (SFT only)	45.09	24.87	26.68	18.72	76.87	48.46

Best F1 in each column is in bold. SkillSpan without retrieval is the matched full split ($N=3569$, STRICT F1 45.09). A previously circulated subset score of 54.52 ($N=1001$) is *not* used. SFT is essential on SkillSpan (53.38 $\rightarrow$ 18.71) and Kompetencer (43.84 $\rightarrow$ 11.56). The backbone with retrieval exceeds the SkillAnchor adapter on Gnehm (41.29 $\rightarrow$ 56.19). On Sayfullina, the same SFT adapter without retrieval reaches 76.87 (16.2% of BIO sequences identical to the SkillAnchor dump); the frozen retrieval setting is strongly harmful on that corpus. SkillAnchor remains the pre-specified configuration, not a per-corpus selection of the highest F1. On Green and Gnehm, disabling retrieval also raises STRICT F1 relative to SkillAnchor.

the reported SFT adapter. (ii) The retrieval-enabled pipeline is stronger on SkillSpan, Kompetencer, and FIJO, whereas the retrieval-disabled pipeline is stronger on Green, Gnehm, and Sayfullina (Sayfullina 25.13 $\rightarrow$ 76.87); because the prompt template also changes, this is not a context-only effect. (iii) CoT is not a reliable STRICT improvement on top of SFT+RAG; the SkillSpan run labeled RAG $k=7$ (44.49) is below the frozen default retrieval run (53.38). (iv) The illegal-BIO rate on stored SkillAnchor predictions is 0.0%; remaining STRICT false positives are span mismatches, not illegal tags.

D. RETRIEVAL-SOURCE DIAGNOSTIC

Table 2 uses the frozen default 5+1+1 train+ESCO retrieval setting. Appendix B reports a matched full- N SkillSpan decode with the archived run label $k=7$ (STRICT F1 44.49), below the default SkillAnchor row; that label should not be read as the default number of retrieved items. Fig. 2 already includes the retrieval-disabled operational variants; retrieval width is not swept in that figure.

Table 4 separates the two sources while retaining the same LoRA adapter family. On SkillSpan, Kompetencer, and FIJO, the concatenated context is stronger than either source alone. On Green, Sayfullina, and Gnehm, however,

concatenation is substantially below the stronger source-only decode, indicating a harmful source interaction rather than a uniformly harmful retriever. A corpus-level routing schedule recorded before the final Gnehm RAG-2 result obtains an unweighted mean of 55.15 and a test-size-weighted mean of 64.57; XLM-R remains higher at 59.26 and 71.77. This routing result is a diagnostic of source selection, not the pre-specified SkillAnchor operating point in Table 2.

E. AUXILIARY VERIFY DIAGNOSTIC

Table 5 asks an auxiliary question separate from SkillAnchor-Gen: whether spans proposed by XLM-R can be selectively edited by Qwen2.5-14B-Instruct with RAG-2 ESCO-by-span. The verifier is invoked only when at least one proposed span has confidence below 0.80 and may KEEP, DELETE, or ADJUST a proposal; it may not add an unconstrained new span. This Instruct-Qwen verifier is not the LoRA adapter used by the Table 2 generator and is not a second SkillAnchor method.

On the frozen SkillSpan clean-development split, RAG-2 editing improves B-started exact-span F1 from 56.47 to 57.32 ($\Delta = +0.85$; bootstrap 95% CI [0.28, 1.46]); the no-RAG and RAG-1 intervals include zero. Because this metric differs from the main seqeval protocol, 57.32 is not compared

TABLE 4: Retrieval-source ablation and corpus-level routing under STRICT F1 (%).

Setting	SkillSpan	Kompetencer	Green	FIJO	Sayfullina	Gnehm	AVG
<i>N</i>	3569	262	335	50	1851	2550	—
RAG-1 (train)	44.21	23.12	41.93	11.43	81.90	72.95	45.92
RAG-2 (ESCO)	43.77	25.25	42.00	12.09	82.38	36.05	40.26
Both (= Table 2 SkillAnchor)	53.38	43.84	23.13	36.36	25.13	41.29	37.19
Without retrieval	45.09	24.87	26.68	18.72	76.87	48.46	40.12
Routed (pre-specified)	53.38	43.84	42.00	36.36	82.38	72.95	55.15
Choice	both	both	RAG-2	both	RAG-2	RAG-1	—

RAG-1 retains five in-domain training neighbors and disables ESCO; RAG-2 disables training neighbors and retains one ESCO occupation plus one ESCO skill entry. “Both” is the frozen 5+1+1 Table 2 configuration. The without-retrieval row also uses the repository’s non-RAG prompt family and is not a fixed-prompt context-only control. The routing schedule was locked before the Gnehm RAG-2 result. Its test-size-weighted mean is 64.57 (Both: 42.17; XLM-R: 71.77). The routed diagnostic is not copied into Table 2.

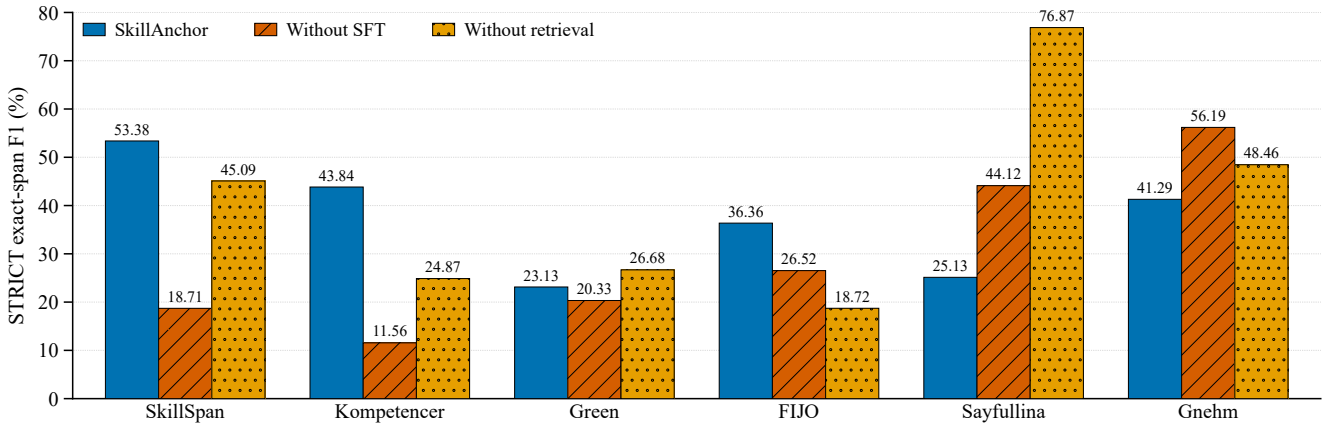


FIGURE 2: Operational SFT and retrieval comparisons under STRICT exact-span F1 (same cells and corpus order as Table 3). Without SFT uses the base 14B backbone with retrieval; without retrieval uses the SkillAnchor adapter with the non-RAG prompt template and no retrieved context, so it is not a context-only intervention. Sayfullina without retrieval (76.87) and Gnehm without SFT (56.19) exceed the pre-specified SkillAnchor row; SkillAnchor is not selected per corpus.

with the Table 2 SkillSpan value. Panel B of Table 5 reports official-test Verify sequeval on all six corpora: SkillSpan 55.33 → 56.81, Kompetencer 43.08 → 45.13, Green 43.07 → 43.29, FIJO 34.82 → 37.07, Sayfullina 91.03 → 91.51, and Gnehm 88.24 → 88.36. Span- Δ bootstrap 95% CIs exclude zero only for SkillSpan ([0.07, 0.99]); the other five include zero (Kompetencer [−3.45, 1.48], Green [−0.65, 1.14], FIJO [−1.33, 2.55], Sayfullina [−0.02, 0.22], Gnehm [−0.48, 0.35]). Kompetencer 45.13 is therefore not a significant span-level gain. The Gnehm clean-split cell remains unpublished. We do not compute a six-corpus Verify mean, and we do not copy 57.32 or 56.81 into Table 2 or replace the XLM-R row.

F. SKILLSPAN EXACT-SPAN ERROR ANALYSIS

Residual SkillSpan STRICT disagreements are often boundary or label-scope disagreements rather than illegal BIO, whose rate is already 0.0%. On the official test set ($N=3569$), 268 gold spans are exact-match true positives for exactly one of XLM-R or SkillAnchor-Gen: 133 are Gen-only and 135 are XLM-R-only. In 136 of these 268 cases (50.7%), the other system overlaps the gold span in tokens, indicating a nested, split, or one-two-token boundary

mismatch; no token overlap occurs for 57 of 133 Gen-only spans (42.9%) and 75 of 135 XLM-R-only spans (55.6%). A seed-42 stratified audit of 50 disagreements (25 from each side) classifies 15 (30%) as plausible complementary misses, 17 (34%) as STRICT boundary mismatches, and 18 (36%) as annotations whose scope is debatable under a narrower transferable-skill interpretation, including personality traits and whole duty clauses. Gen-only hits contain more technical list items and long duty clauses, whereas XLM-R-only hits contain more gold-style soft or personality spans. Because an unfiltered span union is already 0.12 F1 below XLM-R, we do not treat the 133 Gen-only gold matches as an ADD budget for Verify, and we neither retokenize nor relabel the public test. Appendix Table 11 gives a balanced sample.

G. IMPLEMENTATION DETAILS AND REPRODUCIBILITY

LoRA SFT uses rank 8 (see `adapter_config.json` in the supplementary bundle). Decoding is greedy (temperature 0). RAG uses the frozen reported 5+1+1 configuration unless noted: five processed-training sentences, one ESCO occupation entry, and one ESCO skill entry [14]. Construction notes are in the supplementary `rag_index.md`; the historical filename token `rag-default-k3` is a legacy run

TABLE 5: Optional SkillAnchor Verify mode (Hybrid): XLM-R–Instruct–Qwen span editing.

Panel A: frozen SkillSpan clean-development split (B-started exact-span metric)						
Setting	P	R	F1	$\Delta F1$	Bootstrap 95% CI on Δ	
XLM-R encoder	56.87	56.07	56.47	0.00	—	
Rule delete ($c < 0.70$, no Qwen)	60.36	53.93	56.96	+0.49	—	
Verify, no RAG	58.92	54.95	56.87	+0.40	[−0.20, 0.99]	
Verify, RAG-1	58.92	54.30	56.52	+0.05	[−0.74, 0.87]	
Verify, RAG-2	59.15	55.61	57.32	+0.85	[0.28, 1.46]	
Oracle KEEP/DELETE (ceiling)	63.16	56.07	59.41	+2.94	—	

Panel B: completed cross-corpus cells						
Split / setting	SkillSpan	Kompetencer	Green	FIJO	Sayfullina	Gnehm
Official test N	3569	262	335	50	1851	2550
Official-test XLM-R (sequeval)	55.33	43.08	43.07	34.82	91.03	88.24
Official-test Verify RAG-2 (sequeval)	56.81	45.13	43.29	37.07	91.51	88.36
Clean N	3175	250	322	47	1835	—
Clean XLM-R (B-started)	56.47	57.40	41.20	36.97	90.96	—
Clean Verify RAG-2 (B-started)	57.32	57.01	40.93	37.07	90.94	—

This diagnostic is not Table 2. XLM-R proposes spans; Qwen2.5-14B-Instruct applies KEEP/DELETE/ADJUST only when at least one proposed span has confidence below 0.80, and it cannot add a new span. The verifier is not the LoRA adapter used by the Table 2 SkillAnchor generator. RAG-2 retrieves ESCO context by proposed span. Panel A uses a B-started exact-span metric; the same Verify RAG-2 dump scores 54.64 with `evaluate_src.py`, so 57.32 must not be compared with the Table 2 SkillSpan value. The oracle is a ceiling, not a deployable system. Dashes denote unavailable cells; no six-corpus Verify mean is reported.

label rather than the number of inserted items. All XLM-R, GLiNER, and SkillAnchor scores in Table 2 are recomputed with `evaluate_src.py` (sequeval exact-span) from stored prediction JSON (or the `.eval_ner.json` sidecar when the dump’s gold tags are empty). The historical aggregates in Appendix Table 12 are the explicit exception. Appendix F maps each reported SkillAnchor cell to a file. The supplementary bundle contains a manifest, a one-command scorer (`score_tables.py`), frozen prompts, LoRA configs, and SHA-256 hashes of the processed train/test JSON files. An independent reader can regenerate every Table 2 cell and every Table 3 cell from those artifacts without a new decode; the historical prompting rows in Appendix Table 12 are preserved aggregates rather than regenerable cells. Dataset-specific templates in `prompt_template_rag.py` were frozen before the Access re-evaluation and were not retuned against test F1.

H. LIMITATIONS, UNCERTAINTY, AND ETHICS

Every reported F1 is a single decode; we do not provide multi-seed standard deviations. Table 6 reports a 1000-resample sentence bootstrap on the stored SkillAnchor predictions. SkillSpan (53.38, 50.30–56.53) and Sayfullina (25.13, 23.43–26.85) have reasonably tight intervals; Kompetencer and FIJO do not ($N=262$ and $N=50$). Illegal-BIO sentences are 0.0% on all six SkillAnchor files: BIO-only decoding already yields legal tag sequences. Remaining STRICT errors are span mismatches (exact-span false-positive rates 48–78%), not illegal tags. We therefore do not equate a zero illegal-BIO rate with hallucination-free extraction.

The RAG train index contains no extra test sentences beyond the overlap already present in the processed split: unique $RAG \cap test$ strings equal unique $train \cap test$ strings on every corpus. That overlap is not negligible at the record level on SkillSpan (23.1%) and Gnehm (20.3%), because some

overlapping strings repeat. Dropping those records changes stored exact-span F1 by less than one point (SkillSpan 53.38 $\rightarrow$ 53.31; Gnehm 41.29 $\rightarrow$ 40.34), so the Table 2 scores are not an artifact of exact-sentence retrieval. Head-mention recall exceeds tail-mention recall on every corpus; the gap is largest on Sayfullina (32.2 vs. 2.1).

FIJO has only 50 test sentences, so its F1 is unstable under a few span errors. The unweighted macro-average treats all six corpora equally and is not a size-weighted production metric; we also report the N -weighted mean 42.17 in Table 2. Matched STRICT scores for XLM-R and GLiNER are reported in Table 2. SkillSpan and Kompetencer XLM-R models were retrained because the available checkpoints were incomplete; the other XLM-R cells use stored checkpoints, and GLiNER is fine-tuned per corpus.

The study uses public job-ad benchmarks and ESCO. No new human-subject annotation was collected. Extracted spans are intended for labor-market research. They must not be used as the sole input to automated candidate screening, ranking, or hiring. Span errors, head-skill bias (Table 6), train/test string overlap, and uneven ESCO coverage can systematically disadvantage some applicants. Any operational use requires human review of both recalled and missed skills. The checker, when used, only constrains BIO legality; it does not certify factuality, ESCO membership, or fairness.

IV. RELATED WORK

Job-skill extraction. Skill extraction from job advertisements is commonly formulated as span identification or sequence labeling, with public resources differing in language, sector, and skill definition [1], [2]. Nguyen et al. [20] unify the six corpora used here and evaluate in-context LLM prompting; they find that LLMs can handle some syntactically complex mentions while remaining below supervised BIO taggers in aggregate. SkillAnchor builds on that benchmark setting but trains an open 14B extractor, adds a frozen

TABLE 6: CPU diagnostics on stored SkillAnchor predictions (seqeval STRICT F1 with a 1000-resample sentence bootstrap).

Dataset	<i>N</i>	F1	95% CI	Ill.%	Dup uniq (rec%)	Excl. F1	Head R	Tail R
SkillSpan	3569	53.38	50.30–56.53	0.0	177 (23.1)	53.31	66.7	52.4
Kompetencer	262	43.84	29.67–54.79	0.0	10 (4.6)	43.84	100.0*	38.7
Green	335	23.13	19.19–27.00	0.0	13 (3.9)	22.89	27.1	17.3
FIJO	50	36.36	21.56–49.64	0.0	3 (6.0)	35.83	82.4	30.2
Sayfullina	1851	25.13	23.43–26.85	0.0	15 (0.9)	24.89	32.2	2.1
Gnehm	2550	41.29	37.32–45.24	0.0	257 (20.3)	40.34	45.6	30.9

Ill.%: illegal-BIO sentence rate. Dup uniq: unique train/test sentences; rec% is the share of test *records* with those strings. Excl. F1 drops those records. Head/tail: gold mention seen vs. unseen in train. *Kompetencer head has only 12 gold spans.

train-split and taxonomy-definition retrieval layer, and separates SFT and retrieval effects under a regenerable STRICT evaluation.

Supervised and generalist extractors. Multilingual encoders such as XLM-R provide strong token-classification backbones [3]. Generalist systems such as GLiNER and UniversalNER instead condition extraction on entity descriptions or broad distillation data [4], [5]. These model families are not architecturally identical to a BIO-generating LLM, but they can be compared empirically when the split, gold spans, and exact-span scorer are held fixed. We therefore include matched XLM-R and GLiNER rows in Table 2 and avoid treating architectural difference as evidence of superiority.

Generative information extraction. Unified IE and instruction-tuned frameworks map text to structured records across tasks [6]–[8]. Few-shot studies likewise show that large models can complement smaller extractors, but may incur more false positives, latency, and cost when used as stand-alone IE systems [21]. Chain-of-Thought and self-consistency improve reasoning in some settings [22], [23]; they do not imply exact span boundaries, so our CoT variants are treated as prompt ablations rather than a separate model family.

Retrieval augmentation and noisy context. RAG couples a generator with retrieved external evidence [9], but retrieval is not uniformly beneficial. Studies on mathematical reasoning, open-domain question answering, and long-context retrieval show that irrelevant context can distract LLMs, that highly ranked non-answer passages may be more damaging than random documents, and that useful information can be underused when it appears in the middle of a long prompt [10]–[13]. NoiserBench further separates beneficial from harmful retrieval noise rather than treating all noise as one condition [24]. These studies do not directly establish the effect of retrieval noise on BIO skill extraction; they motivate our conservative interpretation of the corpus-dependent RAG ablation and the need for source-level controls.

Taxonomy grounding and output control. Recent labor-market systems use ESCO or related taxonomies for representation learning, multilingual classification, or knowledge-graph matching [14], [25]–[27]. SkillAnchor uses ESCO definition text only as retrieved context for span generation; it does not perform occupation classification or skill-ID linking. Verification-augmented NER and constrained decoding target a different failure surface—the structural or semantic validity of generated outputs [28], [29]. Our optional checker

is narrower still: it tests BIO legality and can trigger a retry, but it does not certify factuality, ontology membership, or fairness.

V. CONCLUSION

We presented SkillAnchor as an open-weight, artifact-mapped framework for span-level skill extraction from job advertisements. Its sole reported method, SkillAnchor-Gen, combines span-anchored LoRA SFT, retrieval from annotated training examples and ESCO definitions, dataset-specific prompting, and BIO-only decoding under one exact-span evaluation contract.

On documented processed splits derived from six public corpora, SkillAnchor-Gen is competitive on SkillSpan and leads the artifact-verified XLM-R/GLiNER comparison only on Kompetencer; XLM-R remains strongest in aggregate. These results position SkillAnchor as a transparent generative baseline for controlled diagnosis rather than a uniformly stronger replacement for supervised encoders.

The component results are more informative than a single average. SFT is important on SkillSpan and Kompetencer but is not uniformly beneficial, and the retrieval-disabled comparison is confounded by its different prompt family. Source-isolated decodes provide the clearest finding: concatenating training examples and ESCO definitions helps on SkillSpan, Kompetencer, and FIJO but underperforms a single source on Green, Sayfullina, and Gnehm. Heterogeneous retrieval sources should therefore be evaluated separately before they are combined.

Checker, corpus-level routing, and XLM-R–Qwen Verify experiments remain auxiliary diagnostics and do not alter the reported SkillAnchor-Gen row. Among the Verify comparisons, only the SkillSpan span-level bootstrap interval excludes zero. The current evidence is also limited by single-decode reporting and the absence of a non-test validation protocol for learned source selection. Future work should evaluate noise-controlled retrieval and routing on held-out development data, report efficiency alongside accuracy, and test whether the observed source interactions generalize beyond the six processed corpora.

APPENDIX A A1–A4 CHECKER VARIANTS

Table 7 reports the classify+CoT+checker stack for the corpora whose complete prediction artifacts are included in the manuscript bundle. The SkillAnchor rows in Tables 2 and 3 include neither a classifier nor a checker; A4 is not a

substitute for SkillAnchor. Partial A-series runs are not mixed with full-test results.

Table 8 summarizes every completed full-test A-series F1 cell available for this revision. Sayfullina A4 is the completed $N=1851$ cell, with STRICT P/R/F1 80.90/80.51/80.70 and skill-level P/R/F1 93.61/93.56/93.59; A2 remains highest in STRICT F1 on that corpus (80.77). Sayfullina A3 stopped at $N=1251$ and remains a dash rather than a partial-decode score. A-series cells are not merged into Table 2.

Table 9 reports a separate diagnostic that enables the BIO checker on the corresponding SkillAnchor adapter without the A1–A4 classifier/CoT stack. The checker lowers SkillSpan F1 but raises Gnehm F1, so it remains an optional module and is not folded into Table 2.

APPENDIX B PROMPT AND K VARIANTS

Table 10 reports dataset-agnostic versus dataset-specific prompts, CoT, and an archived RAG run labeled $k=7$ on the corpora with matched full- N files. These rows complement Table 3 but do not isolate SFT or RAG.

APPENDIX C SKILLSPAN STRICT DISAGREEMENT AUDIT

Table 11 illustrates the balanced manual audit summarized in Section III. The examples are descriptive diagnostics from the fixed public test labels; they are not used to relabel the test set or tune Verify.

APPENDIX D SUPPLEMENTARY ABLATION EVIDENCE

To complement the absolute STRICT F1 values in Table 3 and Fig. 2, Fig. 3 re-expresses each matched ablation as a percentage-point change relative to the pre-specified SkillAnchor configuration. Panel A summarizes corpus-wise component effects, whereas Panel B compares prompt and retrieval-width variants on the corpora with matched full- N files. The SkillSpan row is retained as an anchor to the single-corpus comparison; it is not an additional experiment.

APPENDIX E HISTORICAL PROMPTING RESULTS

Table 12 preserves the GPT-3.5 and Qwen prompting aggregates reported in the prior manuscript version. Moving them out of Table 2 separates the artifact-verified comparison from values that cannot be independently re-scored, without deleting unfavorable evidence. In particular, the historical GPT-3.5 5-shot result on Gnehm is 54.91 F1, above the fixed SkillAnchor result of 41.29. These values are reported for provenance and are not used for bolding, ranking, or aggregate comparisons.

APPENDIX F RESULT PROVENANCE

Each STRICT cell of the SkillAnchor row in Table 2 is recomputed with `evaluate_src.py` (seqeval exact-span) from a stored prediction file. Table 13 lists the adapter used for that file. The SkillSpan prediction has a mislabeled experiment field in file metadata (see the supplementary notes); the tags correspond to SFT, a dataset-specific prompt, and

the frozen reported retrieval configuration. CPU diagnostics on the same predictions (illegal BIO, bootstrap intervals, train/test overlap, head/tail recall) are in Table 6. The supplementary bundle lists every Table 2 path and every Table 3 path in `MANIFEST.csv` and regenerates STRICT F1 with the script `score_tables.py`. Historical prompting rows in Appendix Table 12 are excluded from the manifest because their prediction artifacts are unavailable. Legacy SRICL identifiers may remain in stored artifact filenames; all manuscript-facing labels use SkillAnchor.

TABLE 13: Provenance of the SkillAnchor STRICT row in Table 2. Stored predictions live under `output/` by dataset; the metric is seqeval exact-span.

Dataset	N	F1	Adapter
SkillSpan	3569	53.38	sft_skillspan_skills_ner*
Kompetencer	262	43.84	sft_kompetencer_ner
Green	335	23.13	sft_green_ner
FIO	50	36.36	sft_fiyo_ner
Sayfullina	1851	25.13	sft_sayfullina_skills
Gnehm	2550	41.29	sft_gnehm_ner_new

*The SkillSpan stored file has a mislabeled experiment field; tags correspond to SFT, a dataset-specific prompt, and the frozen reported retrieval configuration (see supplementary notes). Gnehm/Sayfullina runs without SFT are under `output/access_rerun/`. XLM-R and GLiNER predictions are tracked separately from the SkillAnchor provenance rows above.

APPENDIX G REPRODUCIBILITY ARTIFACTS

The supplementary zip is the supplementary folder beside this source; upload it with the PDF. No external DOI is required. The script `score_tables.py` regenerates STRICT F1 for completed cells in Tables 2 and 3 whose manifest status is ok.

Train and test JSON files are the documented processed splits of each corpus; hashes are listed in `splits_sha256.txt`. Prompts are frozen in `prompt_template_rag.py` (not retuned on test F1). LoRA uses rank 8 ($\alpha=16$); see `adapter_config.json` in the bundle. The RAG index uses the processed training split plus ESCO v1.2 entries (`rag_index.md`); the reported context contains five training sentences, one occupation entry, and one skill entry. Test sentences are not indexed. Decoding is greedy (temperature 0), BIO-only. Scoring uses `evaluate_src.py` (seqeval exact-span). The SkillSpan SkillAnchor prediction has a mislabeled experiment field in file metadata (documented in the supplementary notes); the run is SFT with a dataset-specific prompt and the frozen reported retrieval configuration, not the dataset-agnostic-prompt ablation. XLM-R and GLiNER driver scripts and matched STRICT predictions support the corresponding rows in Table 2.

TABLE 7: Full-stack A1–A4 for the corpora whose complete prediction artifacts are included in the manuscript bundle. A1: gating classifier + CoT + checker (retrieval disabled). A2: A1 + the frozen default RAG setting (balanced checker). A3: tighter (high-precision) checker. A4: soft classifier filter + high-precision checker. Partial A-series runs are not mixed with full-test results. The SkillAnchor row in Table 2 has no classifier or checker; do not merge A4 into that row.

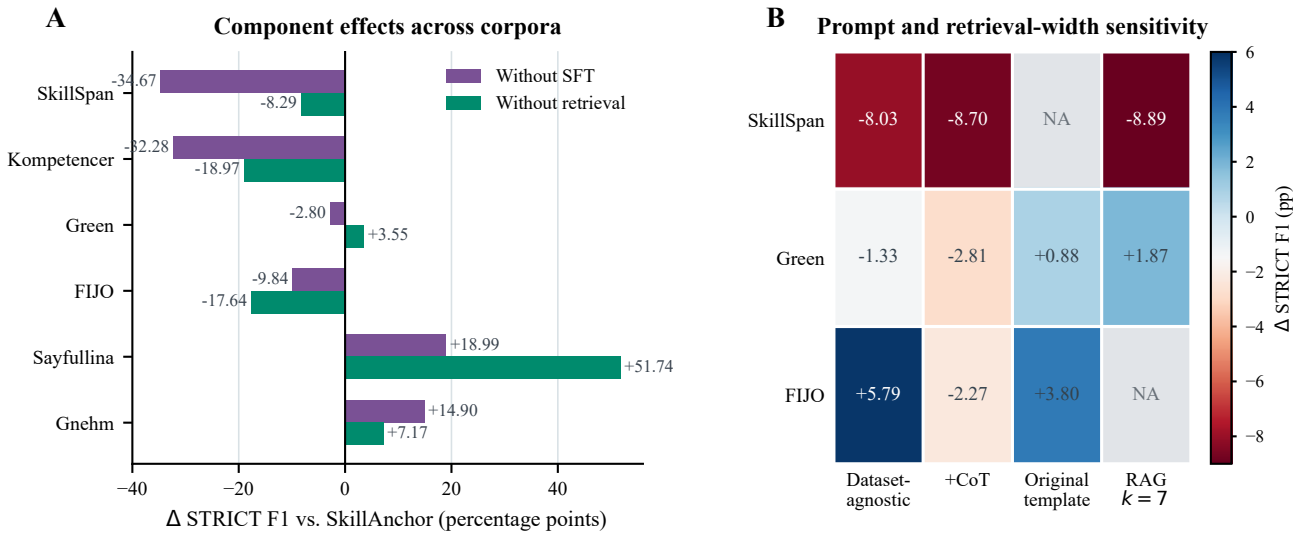
Dataset	Setting	STRICT (segeval)				Skill-level (partial overlap)			
		P	R	F1	$\Delta F1$ vs A1	P	R	F1	$\Delta F1$ vs A1
SkillSpan ($N=3569$)	A1 classify+CoT+checker	34.48	44.00	38.66	—	62.93	80.37	70.59	—
	A2 +RAG	38.14	46.29	41.82	+3.16	63.67	77.34	69.84	−0.75
	A3 tighter checker	35.10	43.72	38.94	+0.28	61.59	76.79	68.35	−2.24
	A4 soft-cls, high-prec. checker	45.93	38.31	41.78	+3.12	80.77	67.43	73.50	+2.91
Gnehm ($N=2550$)	A1 classify+CoT+checker	46.70	45.23	45.95	—	63.38	61.67	62.51	—
	A2 +RAG	46.70	43.66	45.13	−0.82	63.99	60.09	61.98	−0.53
	A3 tighter checker	46.85	44.22	45.50	−0.45	63.38	60.09	61.69	−0.82
	A4 soft-cls, high-prec. checker	47.02	42.54	44.67	−1.28	63.65	57.84	60.60	−1.91

Best F1 per dataset and protocol is in bold. On SkillSpan, A2/A4 raise STRICT F1 vs A1; A4 trades recall for precision and yields the best skill-level F1. On Gnehm, A1 is strongest; adding RAG/checker does not improve STRICT F1. Gnehm A1–A4 uses `sft_gnehm_ner` (not `ner_new` used for SkillAnchor); treat as a matched A1–A4 comparison, not as a replacement for Table 2.

TABLE 8: Cross-corpus coverage of the A1–A4 checker stack under STRICT F1 (%). Only completed full-test cells are reported; incomplete decodes are shown as dashes and are not scored. These settings use the classify+CoT+checker stack and are not substitutes for the checker-free SkillAnchor row in Table 2.

Dataset	N	A1	A2 (+RAG)	A3	A4
SkillSpan	3569	38.66	41.82	38.94	41.78
Kompetencer	262	23.31	26.58	30.19	13.95
Green	335	32.95	34.94	35.02	31.52
FIJO	50	12.72	11.11	16.85	16.85
Sayfullina	1851	80.53	80.77	—	80.70
Gnehm	2550	45.95	45.13	45.50	44.67

Best completed F1 in each dataset row is in bold. Sayfullina A3 stopped at $N=1251$ and is not reported. Its completed A4 cell has STRICT P/R/F1 80.90/80.51/80.70 (+0.17 F1 versus A1) and skill-level P/R/F1 93.61/93.56/93.59; A4 is not merged into Table 2. The detailed SkillSpan/Gnehm precision, recall, and skill-level scores are in Table 7. A-series adapters and prompts are documented separately from the reported SkillAnchor configuration.



Positive values indicate higher STRICT F1 than SkillAnchor (dataset-specific prompt; default retrieval: 5 train + 1 occupation + 1 skill). NA denotes no matched full- N file.

FIGURE 3: Complementary ablation diagnostics under STRICT exact-span F1. (A) Corpus-wise changes after removing SFT or retrieval relative to the pre-specified SkillAnchor configuration. (B) Sensitivity to prompt design and retrieval width on corpora with matched full- N results. Values are percentage-point differences from SkillAnchor; positive values indicate higher STRICT F1. NA denotes that no matched full- N result is available.

TABLE 9: Same-adapter BIO-checker diagnostic under STRICT P/R/F1 (%). The checker is enabled on the corresponding SkillAnchor adapter without the A1–A4 classifier/CoT stack; neither checker result is used in Table 2.

Dataset	Setting	P	R	F1	$\Delta F1$
SkillSpan	SkillAnchor	52.19	54.63	53.38	—
	+BIO checker	38.83	36.02	37.38	−16.00
Gnehm	SkillAnchor	44.08	38.83	41.29	—
	+BIO checker	76.97	39.39	52.12	+10.83

The checker has opposite corpus-level effects: it reduces both precision and recall on SkillSpan, but raises Gnehm precision while leaving recall similar. This diagnostic supports treating the checker as optional rather than folding it into the reported SkillAnchor configuration.

TABLE 10: Prompt and retrieval-width variants under STRICT F1 (%). These rows do not isolate SFT or RAG. SkillSpan cells other than SkillAnchor are full $N=3569$ from the Access re-decode where available.

Setting	SkillSpan	Green	FIJO
N	3569	335	50
SkillAnchor (default 5+1+1 RAG)	53.38	23.13	36.36
Dataset-agnostic prompt	45.35	21.80	42.15
+CoT	44.68	20.32	34.09
Original template	45.00	24.01	40.16
Archived RAG $k=7$ run	44.49	25.00	10.47

SkillSpan dataset-agnostic prompt (F1 45.35), +CoT (F1 44.68), original template (F1 45.00), and the archived run labeled $k=7$ (F1 44.49) are full $N=3569$ re-decodes; the label does not denote the default number of retrieved items. Green/FIJO cells are existing full- N matched-adapter files. FIJO at archived $k=7$ reaches 10.47, below the default 36.36. Original template is the pre-revision instruction text (pm-old), not the reported dataset-specific prompt.

TABLE 11: SkillSpan STRICT disagreements from the balanced manual audit sample.

ID	Gold matched by	Gold span	Other-system span	Audit verdict
520	Gen	work on ETL pipelines	<i>none</i>	true miss
351	Gen	pair programming	<i>none</i>	true miss
2909	Gen	Perform molecular biology techniques	<i>none</i>	true miss
3096	XLM-R	communication skills	<i>none</i>	true miss
3520	Gen	personal motivation	personal motivation drive	boundary
573	Gen	building front-end layer	building front-end layer of our data	boundary
2402	XLM-R	communication and collaborations skills	communication / collaborations (split)	boundary
1391	XLM-R	enthusiasm	infectious enthusiasm	boundary
805	XLM-R	lazy	<i>none</i>	scope-debatable gold
2595	Gen	Provide your patients with a safe and effective course of treatment	<i>none</i>	scope-debatable gold
3427	XLM-R	open-minded	<i>none</i>	scope-debatable gold
1269	XLM-R	sense of humor!	humor!	scope-debatable gold

Gen denotes the reported SkillAnchor generator. The table contains six Gen-only and six XLM-R-only exact matches, with four examples per verdict. “True miss” means that the other system skipped a plausible skill; “boundary” means that it returned the same mention with different extents or splitting; “scope-debatable gold” denotes a trait or duty whose inclusion depends on the annotation convention. These examples diagnose STRICT errors and do not revise the public gold labels.

TABLE 12: Historical prompting results under STRICT span matching (P/R/F1, %). Values are reproduced verbatim from the prior manuscript and were not independently re-scored in this revision.

Method	SkillSpan			Kompetencer			Green			FIJO			Sayfullina			Gnehm			AVG		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
<i>N</i> (test)	3569			262			335			50			1851			2550			—		
zero-shot+specific (GPT-3.5)	3.08	11.94	4.90	1.54	4.76	2.33	7.89	2.10	3.32	21.43	9.68	13.33	5.11	3.42	4.10	5.92	13.80	8.29	6.42	7.62	6.04
5-shot (GPT-3.5)	35.46	29.50	32.21	27.27	25.00	26.09	24.53	8.72	12.87	20.00	9.09	12.50	30.35	36.88	33.20	56.16	53.72	54.91	27.68	27.15	28.63
+kNN (GPT-3.5)	26.96	23.17	24.92	10.55	23.53	22.22 [†]	22.86	18.31	20.88	41.18	31.82	35.90	24.97	30.08	27.29	55.31	48.48	51.67	25.98	29.23	30.48
+kNN+specific (GPT-3.5)	10.89	24.48	15.08	10.55	20.79	14.00	30.02	27.30	28.59	55.31 [‡]	48.48 [‡]	51.67 [‡]	26.60	32.63	29.31	51.89	52.92	52.40	26.47	34.43	31.84
GPT-3.5 Version (+kNN+specific)	9.10	26.60	13.70	11.20	29.20	16.10	30.80	32.80	31.80	43.70	44.70	44.20	29.80	47.30	36.60	29.90	64.80	40.90	25.75	40.90	30.55
Qwen2.5 (5-shot+kNN+specific) [§]	9.26	8.89	9.07	6.10	4.76	5.35	18.94	11.76	14.51	9.47	10.58	10.00	18.44	28.54	22.41	34.95	38.72	36.74	16.19	17.21	16.35

The associated prediction artifacts are unavailable in the current repository, so these aggregates are excluded from the artifact-matched ranking, bolding, and mean comparisons in Table 2. [†]The reported Kompetencer +kNN F1 (22.22) is not the harmonic mean of its reported P/R (approximately 14.57); it is retained only for provenance. [‡]The FIJO +kNN+specific triple is identical to the historical Gnehm +kNN triple and is not treated as independent evidence. [§]The historical Qwen row overlaps a Llama-labeled preprint block and is retained without ranking until its protocol can be reconciled with a complete re-decode. Previously circulated Llama-3-8B and RELAX blocks are omitted because several cells did not correspond to independent predictions.

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